

MM-MTR-012 - Critical Motor Maintenance Manual

Document ID: MM-MTR-012 Revision: 1.8 Effective Date: 5 December 2025 Site: EastRefinery
Assets: Motor M-501, Motor M-502, boiler feed pump motors, other motors on the critical register
Owner: Electrical Maintenance / Rotating Equipment

1. Scope

This manual covers electrical and mechanical maintenance after trips and during planned work on critical motors. It assumes operations have already followed OP-MTR-003 for the immediate response.

2. After-trip maintenance checklist

Do these before anyone argues for a quick restart:

1. Lock out and tag out the motor and the driven equipment.
2. Download or photograph the relay event record and fault codes.
3. Talk to operations about process conditions at the moment of trip.
4. Do insulation resistance tests, and polarisation index where required.
5. Inspect the terminal box for moisture, loose lugs, tracking, or burnt smell.
6. Check coupling, bearings, and whether the train turns by hand.
7. Confirm cooling fan and filter condition.
8. Only then discuss clearance for restart with the shift supervisor.

If the driven machine is seized or heavy to turn, stop calling it "an electrical trip" until rotating has looked at it.

3. Bearing temperature high

1. Check whether the sensor itself is truthful.
2. Confirm grease type and quantity against the lube sheet. More grease is not always kinder.
3. Check alignment and soft foot.
4. Look for coupling issues transferring load into the bearing.
5. Replace bearings if there is wear debris, fretting, or clearance outside limit.

4. Overcurrent and overload events

1. Confirm no mechanical seizure on the driven machine.
2. Ask operations whether density, viscosity, blocked flow, or a closed valve could have loaded the motor.
3. Check phase current balance and supply voltage.
4. Inspect starter contacts and feeder condition.
5. Review whether repeated starts happened in a short window.

5. Ground fault and moisture

1. Do not re-energise until insulation is acceptable.

2. Dry and clean the terminal box.
3. Inspect cable entries and drain plugs.
4. Look for washdown or rain hood problems on outdoor motors.

6. Related asset order after a motor trip

When time is limited, inspect in this order:

1. Driven machine
2. Coupling
3. Motor bearings and cooling
4. MCC protection and feeder
5. Process valves that can overload the motor on start

For boiler feed pump motors, include strainer DP and recirculation / discharge valve positions in the same investigation pack.

7. Documentation expected in the work order

- Trip code and time
- Tests performed and results
- Parts replaced
- Whether operations recommendations matched what you found
- Clear statement of restart readiness: ready / not ready / ready with conditions

8. Related documents

- OP-MTR-003
- TG-MTR-022
- SI-GEN-030
- OP-BFP-001 and MM-BFP-010 when the motor drives a BFP